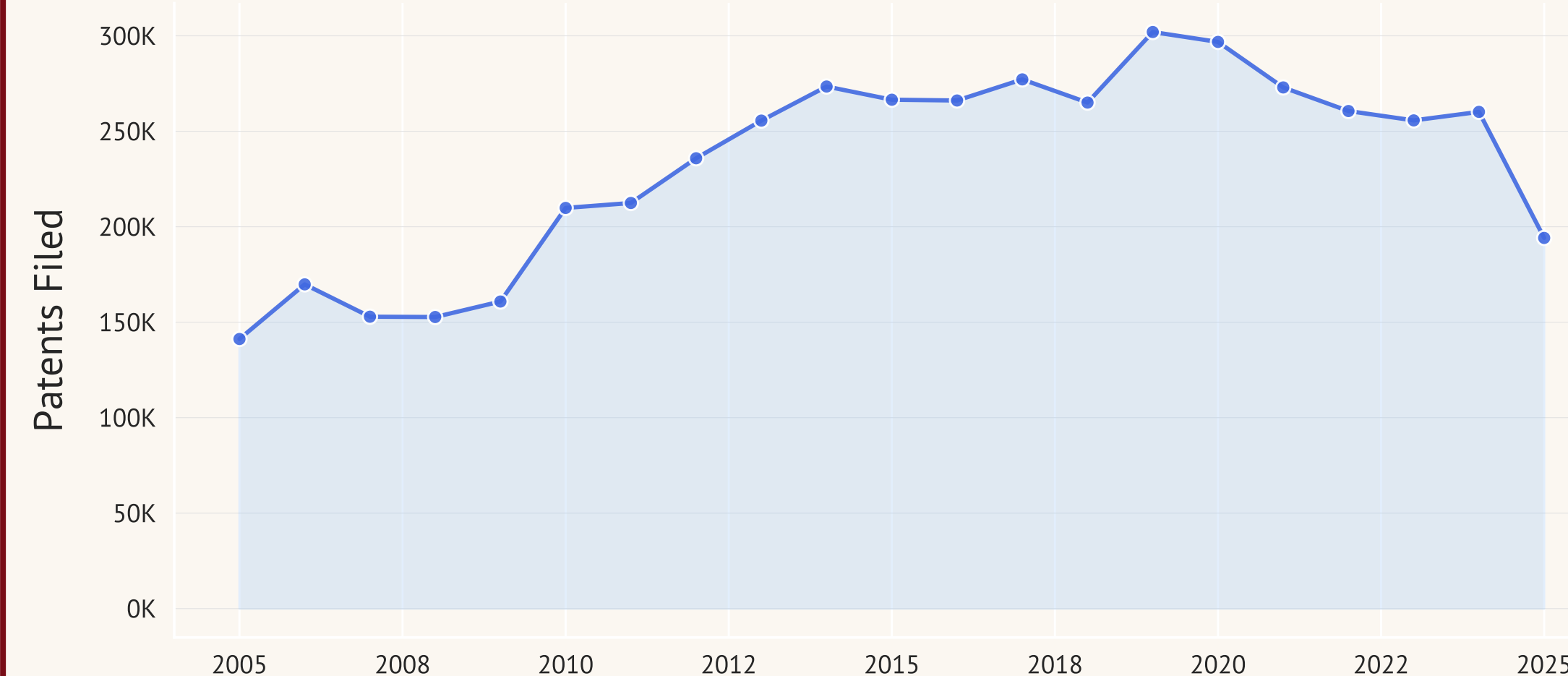


Introduction and Motivation

- Analyzing 122M citation relationships across 6.2M patents filed since 2005.
- All patents are filed with the United States Patent and Trademark Office.
- Patents invented overseas can later be filed with the USPTO.
- Patents cite earlier "prior art" to establish novelty and non-obviousness, the legal bar for patentability.
- Each citation is a directed edge from citing to cited patent, forming the network we analyze.

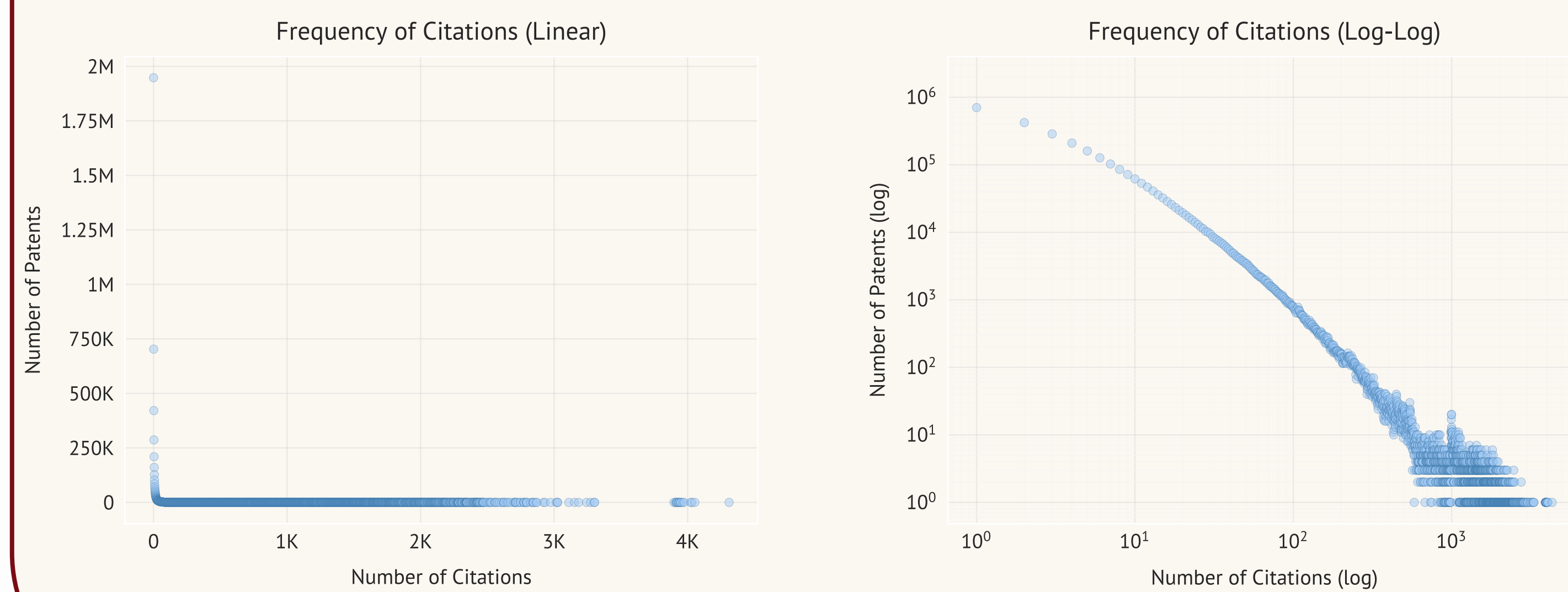
Yearly Patents Count 2005–2025



Top 5 Most Cited Patents

Title	Year	Filing Company & State	Citations
Semiconductor device	2007	Hewlett-Packard. Corvallis, OR	4,024
Combined binary oxide semiconductor device	2007	Hewlett-Packard. Corvallis, OR	3,974
Methods of making thin film transistors comprising ZnO-based semiconductor materials	2008	Kodak. Rochester, NY	3,950
Transistor using an isovalent semiconductor oxide as the active channel layer	2008	Hewlett-Packard. Corvallis, OR	3,939
High-electron mobility transistor with ZnO	2006	Cermet Inc. Smyrna, GA	3,918

In-Degree Distribution



Top 5 Most Cited CPC Subclasses

CPC Subclass	Patents	Citations
Diagnosis; Surgery; Identification	141,639	6,890,035
Electrical Digital Data Processing	466,234	3,760,393
Transmission Of Digital Information	248,350	2,260,239
Information & Communication Technology	106,740	1,371,415
Filters Implantable Into Blood	52,449	1,125,595
Vessels; Prostheses		

Top 10 Most Cited States

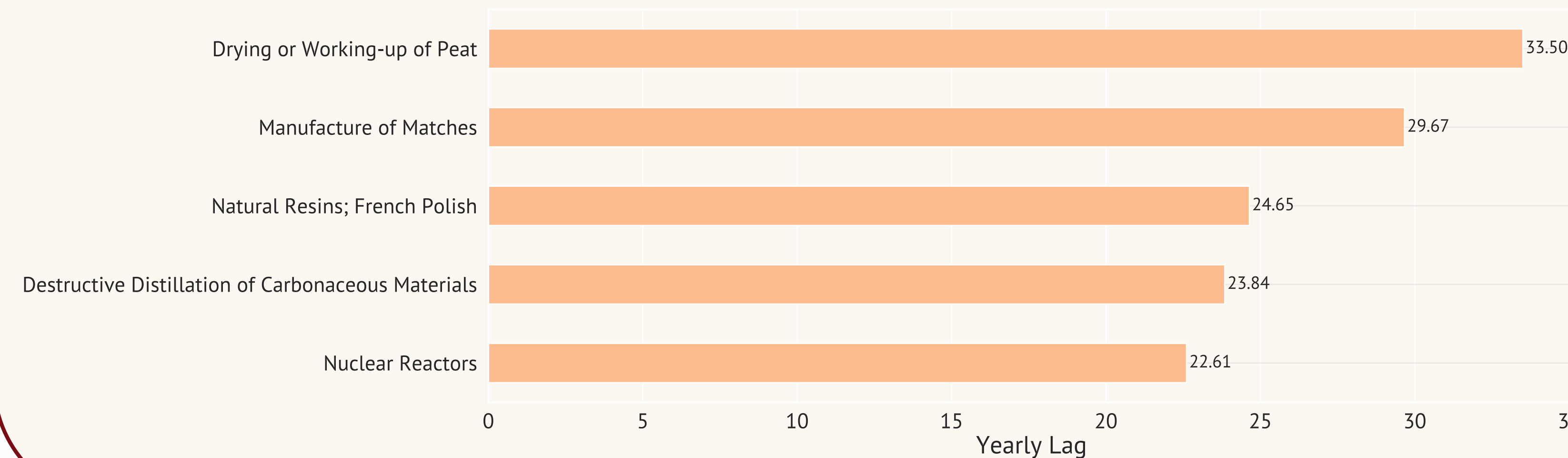
Rank	State	Patents	Citations
1	CA	943,125	32,217,236
2	OH	108,836	10,095,896
3	MA	192,652	5,570,915
4	WA	167,706	5,279,899
5	NY	236,491	4,581,317
6	TX	259,876	4,534,645
7	CT	72,055	4,277,071
8	MN	104,091	3,316,599
9	MI	138,378	3,151,251
10	NJ	137,517	3,005,566

Measuring Citation Time Lag

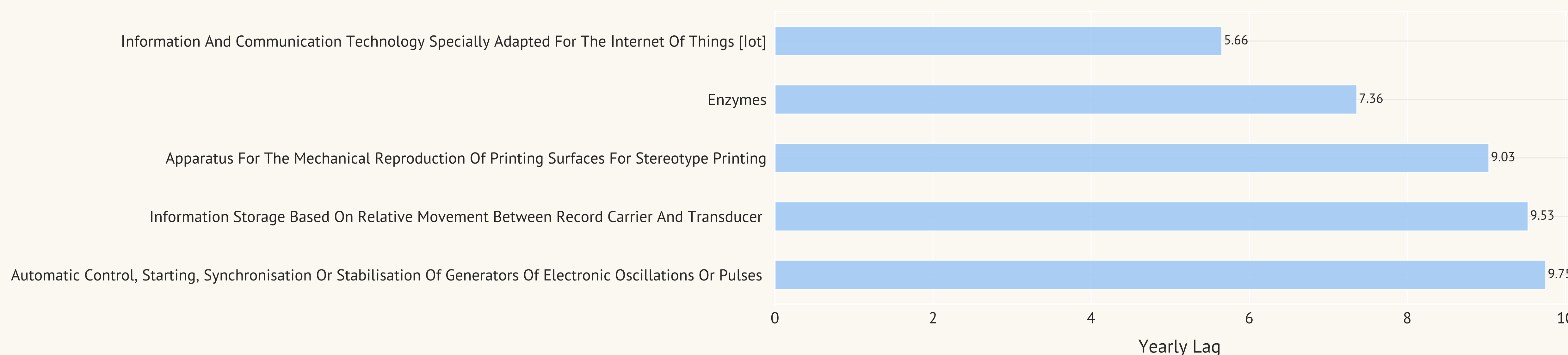
For each patent P filed in year Y_P
For each patent Q filed in year Y_Q that **cites** patent P
Calculate time lag ΔT as $\Delta T = Y_Q - Y_P$

Observe time lag variations across years and different CPC subclasses

Top 5 CPC Subclasses with the Highest Time Lags



Top 5 CPC Subclasses with the Smallest Time Lags



Main Takeaways

In-degree distribution

- The network follows a power-law distribution with a heavy tail.
- A few patents accumulate hundreds to thousands of citations, while most are only cited a handful of times, consistent with preferential attachment in innovation.

Growth trajectory

- The 2010-2014 surge in patents filed reflects the smartphone boom, the rise of e-commerce platforms.
- America Invents Act*: Signed by Obama in 2011, transitioned the U.S. to a first-to-file system by 2013, driving the filing surge.

Ohio's outsides influence

- Ohio ranks 9th in filings but 2nd in total citations received, with ~93 citations per patent.
- Ohio's patents are cited at nearly 3x the rate of California's, indicating a concentration of highly influential inventions.

Citation lag by field

- Emerging tech like IoT (5.7 yrs) and enzymes (7.4 yrs) cite recent prior art.
- Slow-moving industrial fields like peat processing (33.5 yrs) and nuclear reactors (22.6 yrs) draw on decades-old foundations, revealing how innovation speed varies dramatically across domains.